Chapter 1 Homework

Prerequisites

Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Give exact answers. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Then complete the four self-evaluation questions in Blackboard, attach a photo of the one page you are proudest of, and retain all your written work.

Arithmetic and algebra essentials

1. Evaluate.

$$-3^2 + 2(-4+1)^2$$

2. Evaluate. Write your answer as a reduced fraction.

$$\frac{\frac{2}{3} - \frac{1}{4}}{\frac{5}{6}}$$

3. Evaluate when x = -2.

$$2x^2 - 3x + 1$$

4. Simplify. Name the property you use to remove the parentheses.

$$3(2x-5) - 2(x+4)$$

Exponent laws

For 5-6, assume x and y are nonzero. Simplify using only positive exponents.

5. Simplify.

$$\frac{(3x^{-2}y^3)^2}{9xy^{-1}}$$

6. Simplify.

$$\left(\frac{2x}{3y}\right)^{-2}$$

Radicals and rational exponents

7. Simplify.

$$\sqrt{48} + \sqrt{27}$$

8. Simplify for every real value of x, including negative values.

$$\sqrt{25x^2}$$

9. Evaluate each expression exactly.

a) $16^{3/4}$ b) $27^{-2/3}$

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Continue on your own paper

Polynomial operations

10. Simplify. State the degree and leading coefficient of the result.

$$(4x^2 - 3x + 2) - (x^2 + 5x - 6)$$

11. Expand and combine like terms.

$$(2x - 3)(x + 4)$$

12. Expand and combine like terms.

$$(x - 5)^2$$

Factoring

For 13-16, factor completely over the integers.

13. Factor completely.

$$12x^3 - 27x$$

14. Factor completely.

$$x^2 - x - 20$$

15. Factor completely.

$$6x^2 + x - 2$$

16. Factor completely.

$$x^3 + 3x^2 - 4x - 12$$

Rational expressions

For 17-19, simplify and list every excluded value from the original expression.

17. Simplify.

$$\frac{x^2 - 16}{x^2 - x - 12}$$

18. Multiply and simplify.

$$\frac{x^2 - 9}{x^2 - 4} \cdot \frac{x + 2}{x - 3}$$

19. Subtract and simplify.

$$\frac{2}{x - 1} - \frac{1}{x + 1}$$

20. A student says that (x + 5)/x simplifies to 5 by canceling the x terms. Explain the error, then use x = 1 to show why the proposed simplification fails.